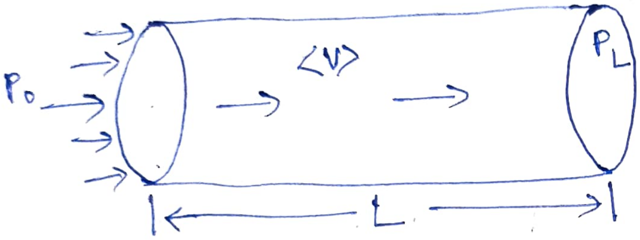
UNIT-5: INTERFACE TRANSPORT

Friction Factors

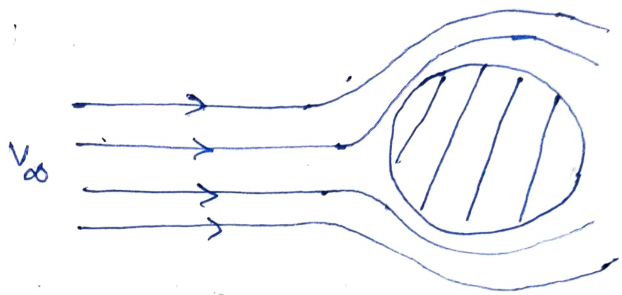
Assumptions: ① Steady state ② Incompressible fluid.

Consider 2 configurations (type of flow):

ⓐ Flow in a straight pipe or channel



ⓑ Flow around submerged object



Force exerted by the fluid on solid surfaces:

$$F_{f \to s} = F_s + F_k$$

where, F_s = Force that would be exerted by the fluid even if it were stationary.

F_k = Additional force associated with the motion of fluid.

It has been observed experimentally that

$$F_k \propto AK$$

$$\Rightarrow F_k = f\,AK \qquad ①$$

where, A = characteristic Area

K = Kinetic Energy per unit volume

f = Friction factor

ⓐ Flow in Tubes/Pipe

$$F_k = (P_0 - P_L)\,\pi R^2 \qquad ②$$

$$A = 2\pi R L \qquad \& \qquad K = \frac{1}{2}\rho\langle v\rangle^2$$

$$\therefore \; F_k = f\,(2\pi R L)\left(\tfrac{1}{2}\rho\langle v\rangle^2\right) = \pi R^2 (P_0 - P_L)$$

$$\Rightarrow f = \frac{\pi R^2}{2\pi R L}\;\frac{P_0 - P_L}{\frac{1}{2}\rho\langle v\rangle^2} = \frac{R}{2L}\;\frac{P_0 - P_L}{\frac{1}{2}\rho\langle v\rangle^2}$$

$$= \frac{R\,(P_0 - P_L)}{L\,\rho\langle v\rangle^2} \qquad (3)$$

For Laminar flow,

$$\langle v\rangle = \frac{R^2 (P_0 - P_L)}{8\mu L} \Rightarrow P_0 - P_L = \frac{8\mu L\langle v\rangle}{R^2} \qquad (4)$$

substituting

$$f = \frac{8\mu R L\langle v\rangle}{L\rho\langle v\rangle^2 R^2} = \frac{8\mu}{\rho R\langle v\rangle} = \frac{16\mu}{D\langle v\rangle\rho} = \frac{16}{Re}$$

where $Re = \dfrac{D\langle v\rangle\rho}{\mu}$

Eq. (2) appears from

$$F_k = \tau_{rz}\Big|_R (2\pi R L) = \frac{(P_0 - P_L)R}{2L}(2\pi R L)$$

$$= (P_0 - P_L)\,\pi R^2$$

Eq. (4) $\Rightarrow$

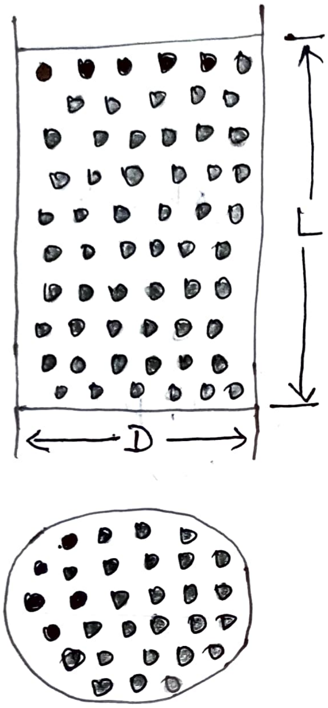
$$\frac{P_0 - P_L}{L} = \frac{8\mu\langle v\rangle}{R^2} = \frac{8\mu\langle v\rangle}{\left(\frac{D}{2}\right)^2}$$

$$\Rightarrow \frac{\Delta P}{L} = \frac{32\mu\langle v\rangle}{D^2}$$

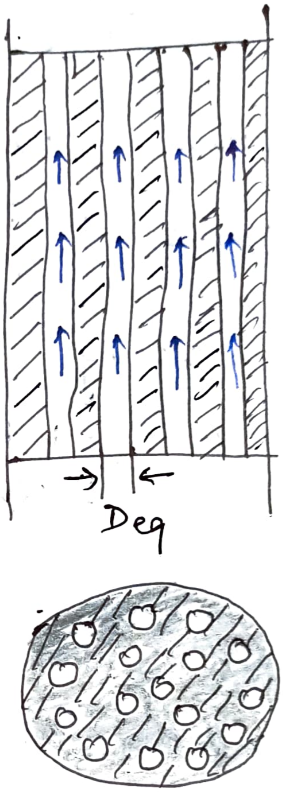
Friction Factors for Packed Columns (ERGUN Eq)

Consider a packed column of height L & cross-sectional area S_0.

Actual Column



simplified column



D = Diameter of column

L = Height of column

S_0 = cross-sectional area of column (Total) $= \frac{\pi}{4} D^2$

Assumptions:

① All the particles are of same size and shape.

~~② The packed bed is analogous to "n" number of parallel~~

② The empty space in the column is analogous to "n" number of hollow parallel channels of uniform cross-section. ③ Porosity of particles is neglected.

n = Number of (Hypothetical) channels

D_{eq} = Equivalent diameter of channels

S = cross-sectional area available for fluid flow.

$= \left(\frac{\pi}{4} D_{eq}^2\right) n$

① superficial velocity (V_0)

The velocity of fluid in empty column.

For an incompressible fluid,

$$SV = S_0 V_0 \quad ①$$

where, V = Actual fluid velocity

V_0 = superficial velocity

② Porosity (voidage or void fraction), ϵ

$$\epsilon = \frac{\text{Empty volume of column}}{\text{Total volume}}$$

Total volume = Solid volume (occupied by particles) + Empty volume (voids)

$$\Rightarrow (1-\epsilon) = \frac{\text{Solid volume}}{\text{Total volume}}$$

③ Sphericity (ϕ)

D_p = Nominal Particle Diameter

= Diameter of sphere of equal volume as the particle

$$\phi = \frac{\text{Surface Area of Sphere of diameter } D_p}{\text{Surface Area of Particle}}$$

Total volume of column = $S_0 L$

Empty volume (void) = $SL = \epsilon S_0 L$

$$\therefore \epsilon = \frac{SL}{S_0 L} = \frac{S}{S_0} \quad ②$$

Eq. ① can be written as

$$V = \frac{S_0}{S} V_0 = \frac{V_0}{\epsilon} \quad ③$$

Spherisity can also be defined as

$$\phi = \frac{(\text{Surface Area}/\text{volume})_{\text{sphere}}}{(\text{surface Area}/\text{volume})_{\text{particle}}}$$

$$\left(\frac{\text{Surface Area}}{\text{Volume}}\right)_{\text{sphere}} = \frac{\pi D_p^2}{\frac{\pi}{6} D_p^3} = \frac{6}{D_p}$$

$$\Rightarrow \phi = \frac{6/D_p}{\left(\frac{\text{Surface Area}}{\text{volume}}\right)_{\text{particle}}}$$

$$\Rightarrow \left(\frac{\text{Surface Area}}{\text{volume}}\right)_{\text{particle}} = \frac{6/D_p}{\phi} = \frac{6}{\phi D_p} \quad (4)$$

Surface Area of "n" channels = Total surface Area of particles.

$$\Rightarrow (\pi D_{eq} L) n = \left(\frac{\text{Surface Area}}{\text{volume}}\right)_{\text{particle}} S_o L (1-\epsilon)$$

$$\Rightarrow \pi D_{eq} L n = \left(\frac{6}{\phi D_p}\right)(1-\epsilon) S_o L = \frac{6(1-\epsilon) S_o L}{\phi D_p}$$

$$\Rightarrow \cancel{D_{eq} =} \pi D_{eq} L n = \frac{6(1-\epsilon) S_o L}{\phi D_p} \quad (5)$$

Volume of "n" channels = Empty volume (voids).

$$\Rightarrow \left(\frac{\pi}{4} D_{eq}^2 L\right) n = \epsilon S_o L \quad (6)$$

Divide eq. (6) by eq. (5)

$$\frac{(6)}{(5)} \Rightarrow \frac{\frac{\pi}{4} D_{eq}^2 L n}{\pi D_{eq} L n} = \frac{\epsilon S_o L \phi D_p}{6(1-\epsilon) S_o L}$$

$$\Rightarrow D_{eq} = \frac{4 \phi D_p}{6} \frac{\epsilon}{1-\epsilon} = \frac{2}{3} \phi D_p \left(\frac{\epsilon}{1-\epsilon}\right) \quad (7)$$

(a) For Low Re (Laminar Flow)

$$\frac{\Delta P}{L} = \frac{32 \mu V}{D_{eq}^2} = \frac{32 \mu \frac{V_0}{\epsilon}}{\left[\frac{2}{3} \phi D_p \left(\frac{\epsilon}{1-\epsilon}\right)\right]^2}$$

$$= \frac{72 \mu V_0 (1-\epsilon)^2}{\phi^2 D_p^2 \epsilon^3} = \frac{72}{\phi^2} \frac{\mu V_0}{D_p^2} \frac{(1-\epsilon)^2}{\epsilon^3}$$

$$\Rightarrow \boxed{\frac{\Delta P}{L} = \frac{150 \mu V_0 (1-\epsilon)^2}{\phi^2 D_p^2 \epsilon^2}} \qquad (8)$$

This is KOZENY CARMAN EQ.

(b) For High Re (Turbulent Flow)

$$\frac{\Delta P}{L} = \frac{2 f \rho V^2}{D_{eq}} = \frac{2 f \rho V_0^2}{D_{eq} \epsilon^2} \qquad \text{[Using Eq. (3)]}$$

$$= \frac{2 f \rho V_0^2 (1-\epsilon)}{\frac{2}{3} \phi D_p \epsilon^3} = \frac{3 f \rho V_0^2 (1-\epsilon)}{\phi D_p \epsilon^3}$$

$$\Rightarrow \boxed{\frac{\Delta P}{L} = \frac{1.75 \rho V_0^2 (1-\epsilon)}{\phi D_p \epsilon^3}} \qquad (9) \qquad [\because 3f = 1.75]$$

This is Burke Plummer Eq.

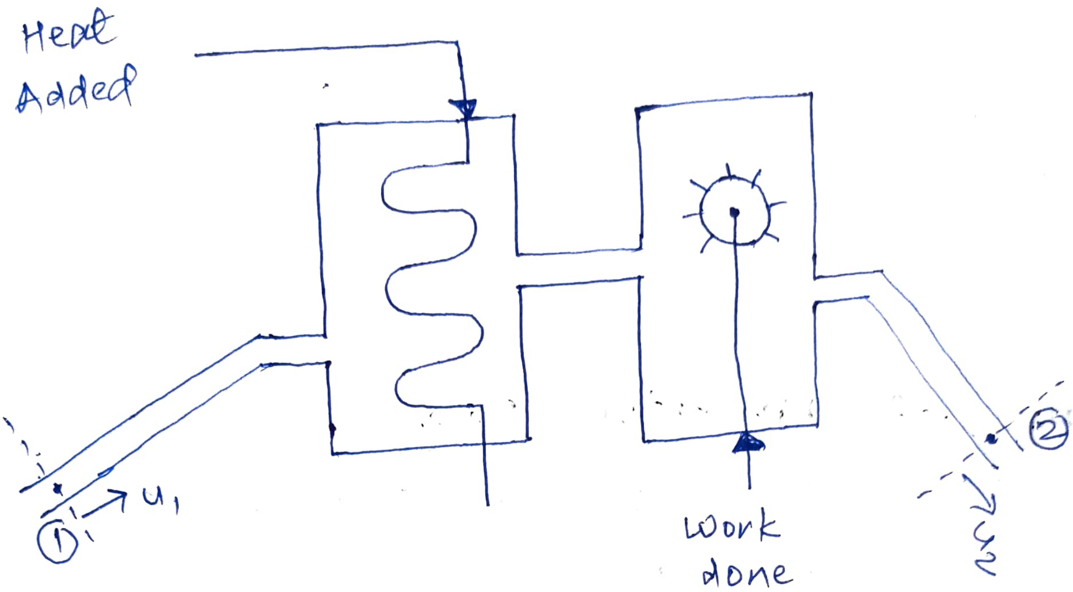
For the entire range of Re, eq. (8) & (9) can be added

$$\boxed{\frac{\Delta P}{L} = \frac{150 \mu V_0 (1-\epsilon)^2}{\phi^2 D_p^2 \epsilon^2} + \frac{1.75 \rho V_0^2 (1-\epsilon)}{\phi D_p \epsilon^3}} \qquad (10)$$

This is ERGUN EQ.

Macroscopic Balance

Consider the following system



Let the fluid enters the system at section ① and leaves at ②.

S_1 & S_2 = cross-sectional area of inlet and outlet pipe at ① and ②

ρ_1 & ρ_2 = density of fluid at ① and ②.

V_1 & V_2 = Average velocity of fluid at ① & ②.

u_1 & u_2 = Unit vectors in the direction of flow at ① & ②.

P_1 & P_2 = Pressures at ① & ②.

w_1 & w_2 = mass flowrates at ① & ②.

$w_1 = \rho_1 V_1 S_1$ and $w_2 = \rho_2 V_2 S_2$

ⓐ Macroscopic Mass Balance

Let m_{total} = mass of fluid contained in the system between ① & ②.

$\Delta(\text{variable}) = (\text{variable})_2 - (\text{variable})_1$

Unsteady state:

$$\frac{d}{dt} m_{total} = w_1 - w_2 = -\Delta w$$
$$= \rho_1 v_1 S_1 - \rho_2 v_2 S_2$$

Steady state:

$$\frac{d}{dt} m_{total} = 0$$

$$\Rightarrow \quad w_1 = w_2 \Rightarrow \rho_1 v_1 S_1 = \rho_2 v_2 S_2$$

(b) Macroscopic Momentum Balance.

Let P_{total} = Net momentum contained in the system between (1) & (2)

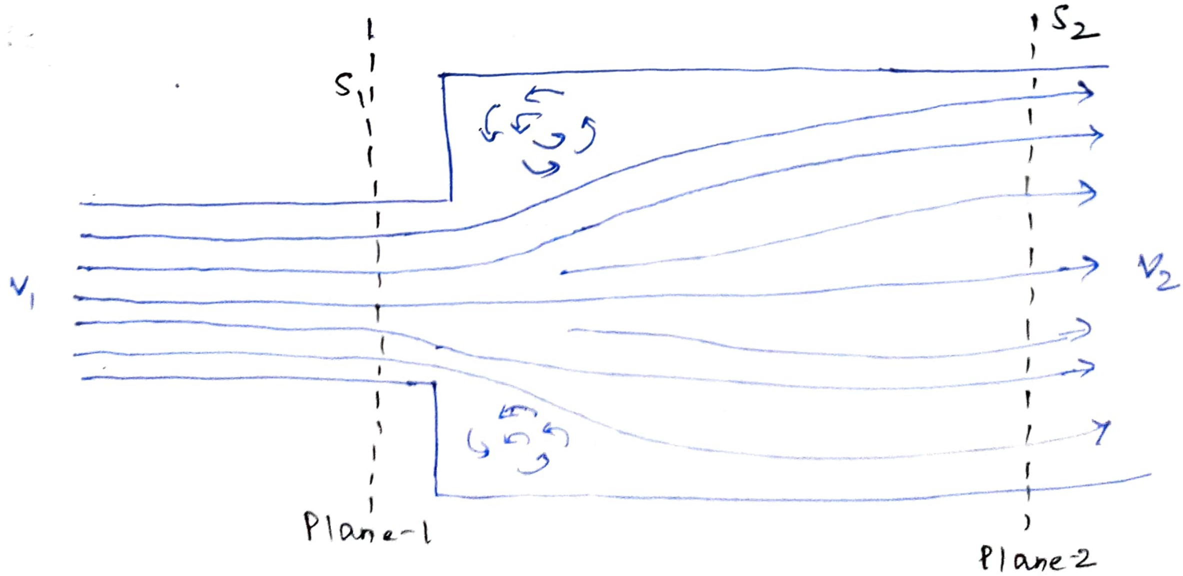
Unsteady state:

$$\frac{d}{dt} P_{total} = (v_1 w_1 - v_2 w_2) + (p_1 S_1 - p_2 S_2) + F_{s\to f} + m_{tot} g$$
$$= -\Delta vw - \Delta pS + F_{s\to f} + m_{tot} g$$
$$= -\Delta(vw + pS) + F_{s\to f} + m_{tot} g$$
$$= -\Delta(\rho v^2 S + pS) + F_{s\to f} + m_{tot} g$$

Steady state: $\frac{d}{dt} P_{total} = 0$

$$\Rightarrow \quad F_{f\to s} = -\Delta(vw + pS) + m_{tot} g$$

(A) Pressure Drop in Sudden Enlargement



Let $\frac{S_1}{S_2} = \beta$

S_1 & S_2 = cross-sectional area of plane 1 & 2

V_1 & V_2 = Average velocity through S_1 & S_2

(a) Mass Balance (Steady State)

$$w_1 = w_2 \Rightarrow \rho V_1 S_1 = \rho V_2 S_2 \Rightarrow V_1 S_1 = V_2 S_2$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{S_2}{S_1} = \frac{1}{\beta} \qquad (3)$$

(b) Momentum Balance

$$F_{f\to s} = -\Delta(pS + vw) = P_1 S_1 + V_1 w_1 - P_2 S_2 - V_2 w_2 \qquad (4)$$

Force over surface $S_2 - S_1$ is

$$F_{f\to s} = -P_1(S_2 - S_1) = P_1 S_1 - P_1 S_2 \qquad (5)$$

Equating (4) & (5)

$$P_1 S_1 - P_1 S_2 = P_1 S_1 + V_1 w_1 - P_2 S_2 - V_2 w_2$$

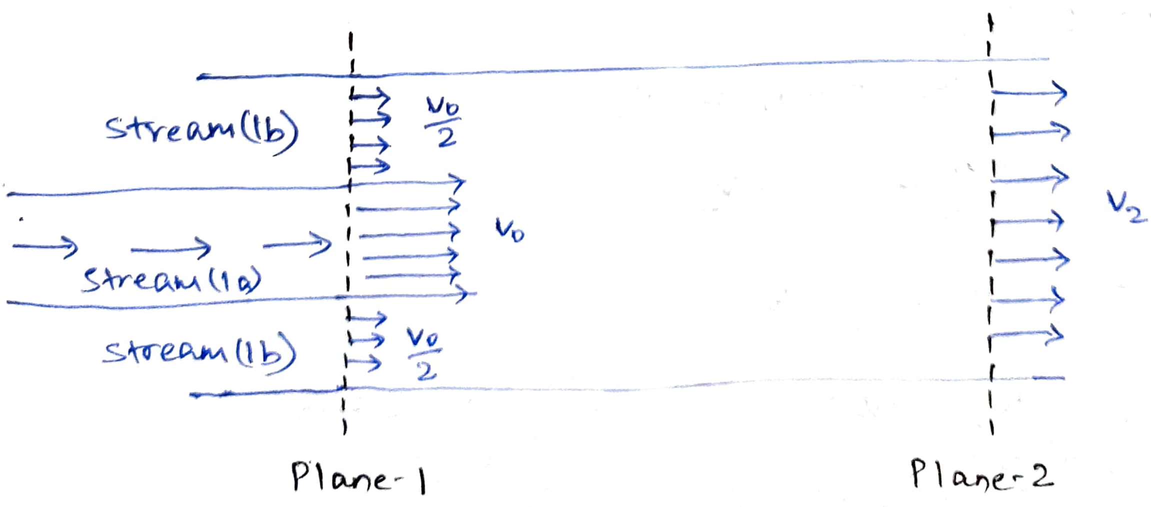
$$\Rightarrow (P_2 - P_1) S_2 = w_2 (V_1 - V_2) \qquad (\text{as } w_1 = w_2)$$

$$\Rightarrow (P_2 - P_1) S_2 = \rho V_2 S_2 (V_1 - V_2)$$

$$\Rightarrow P_2 - P_1 = \rho V_2 (V_1 - V_2) = \rho V_2^2 \left(\frac{V_1}{V_2} - 1\right)$$

$$\Rightarrow \boxed{\Delta p = \rho V_2^2 \left(\frac{1}{\beta} - 1\right)}$$

(B) Liquid-Liquid Ejector



smaller Tube discharges into larger tube containing same fluid at plane-1. Subsequently, these two streams ~~mix~~ get mixed in the downstream,

stream-1a = stream in smaller tube

stream-1b = stream in Larger tube

$V_{1a} = V_0$ = Average velocity of stream-1a,

$V_{1b} = \frac{V_0}{2}$ = Average velocity of stream-1b

S = Cross-sectional area of Larger tube

$\frac{S}{3}$ = Cross-sectional area of smaller tube

V_2 = velocity of mixed stream

Stream-1b flows through annular region with cross-sectional area

$$= S - \frac{S}{3} = \frac{2S}{3}$$

Mass Flowrates :

$$w_{1a} = \rho \frac{V_0}{1} \frac{S}{3} = \frac{\rho V_0 S}{3}$$

$$w_{1b} = \rho \frac{V_0}{2} \frac{2S}{3} = \frac{\rho V_0 S}{3} = w_{1a}$$

$$w_2 = \rho V_2 S$$

① Mass Balance

$$w_{1a} + w_{1b} = w_2$$

$$\Rightarrow \frac{2\rho V_0 S}{3} = \rho V_2 S$$

$$\Rightarrow V_2 = \frac{2}{3} V_0$$

Also, $w_2 = \rho V_2 S = \rho \left(\frac{2}{3} V_0\right) S = 2 w_{1a}$

$$\Rightarrow w_{1a} = w_{1b} = \frac{w_2}{2}$$

(b) Momentum Balance

Assumption: $F_{f \to s} = 0$

$\Rightarrow -\Delta(pS + vw) = 0$

$\Rightarrow P_1 S + v_{1a} w_{1a} + v_{1b} w_{1b} - P_2 S - v_2 w_2 = 0$

$\Rightarrow (P_2 - P_1) S = v_{1a} w_{1a} + v_{1b} w_{1b} - v_2 w_2$

$= v_0 w_{1a} + \frac{v_0}{2} w_{1a} - \frac{2}{3} v_0 (2 w_{1a})$

$= \left(1 + \frac{1}{2} - \frac{4}{3}\right) v_0 w_{1a} = \frac{v_0}{6} w_{1a}$

$\Rightarrow (P_2 - P_1) S = \frac{\rho v_0^2 S}{18}$

$\Rightarrow \boxed{\Delta p = \frac{\rho v_0^2}{18}}$

Nov. 2023

Q. 6 (b). Given,

Diameter, $D = 3\,cm \Rightarrow R = 1.5\,cm = 0.015\,m$

$\rho = 935\ kg/m^3$

$\mu = 1.95\ cP = 0.0195\ P = 0.00195\ Pa.s$

$f = 0.0063$

$Q = 1.1\ \frac{L}{s} = 0.0011\ m^3/s$

Cross-sectional Area, $A = \pi R^2 = 7.0686 \times 10^{-4}\ m^2$

Average velocity, $v = \frac{Q}{A} = \frac{0.0011}{7.0686 \times 10^{-4}} = 1.5562\ m/s$

We know that

$\frac{\Delta P}{L} = \frac{f \rho v^2}{R} = \frac{0.0063 \times 935 \times 1.5562^2}{0.015}$

$= 951.02\ Pa/m$